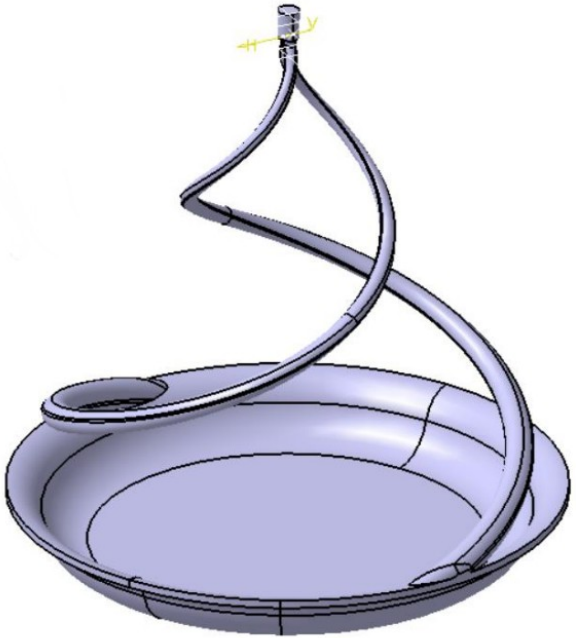


Fabrication du mangeoire bas

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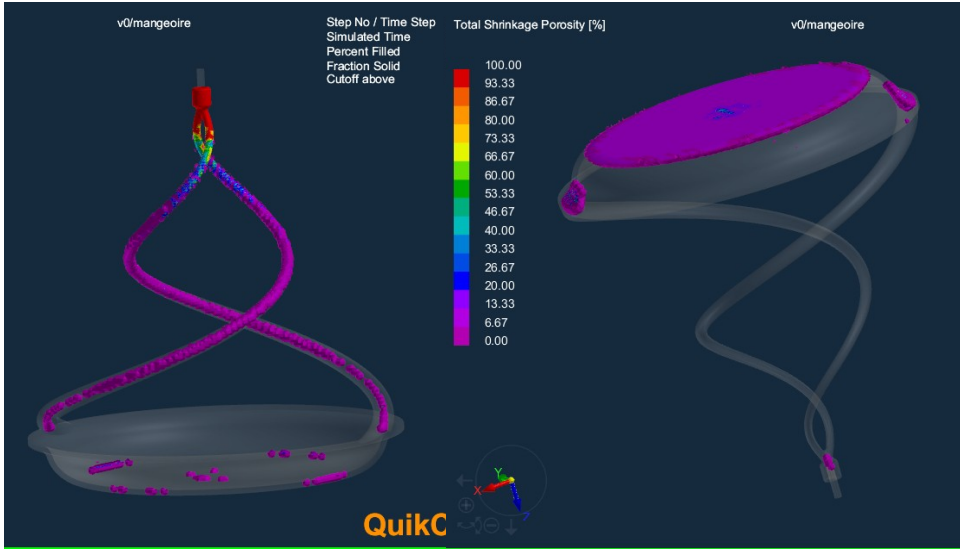
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Sommaire

- Sens de coulée et géométrie
- Dimensionnement des masselottes
- Dimensionnement du système de remplissage
- Modélisation du moule
- Impression du moule et coulée
- Pièce

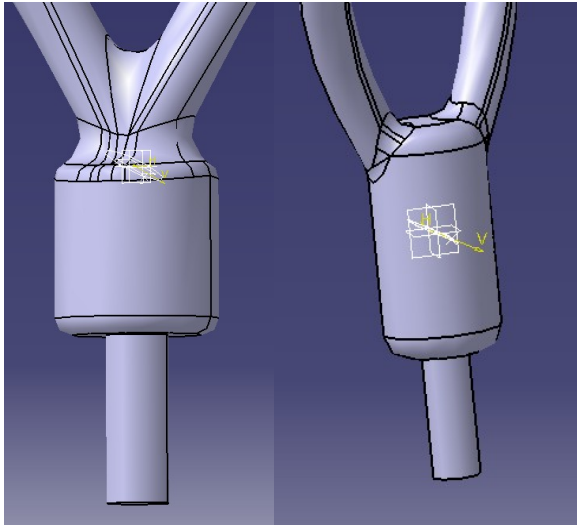
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Sens de coulée et géométrie



Sens retenu

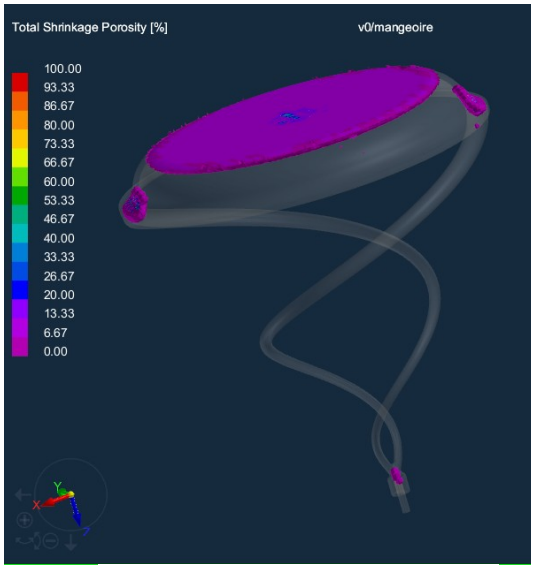
Allongement du diamètre pour accueillir une masselotte



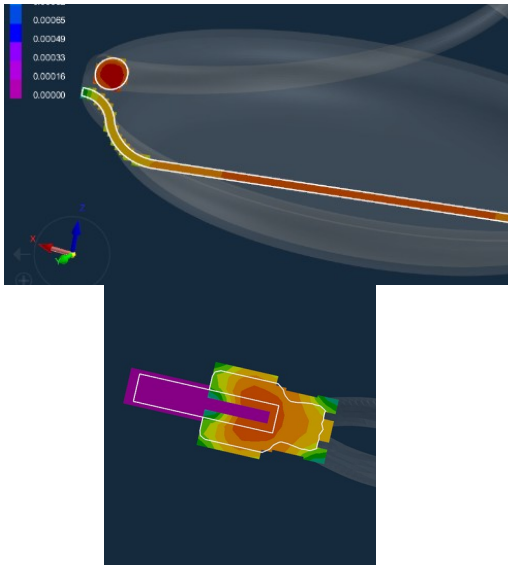
Dimensionnement des masselottes

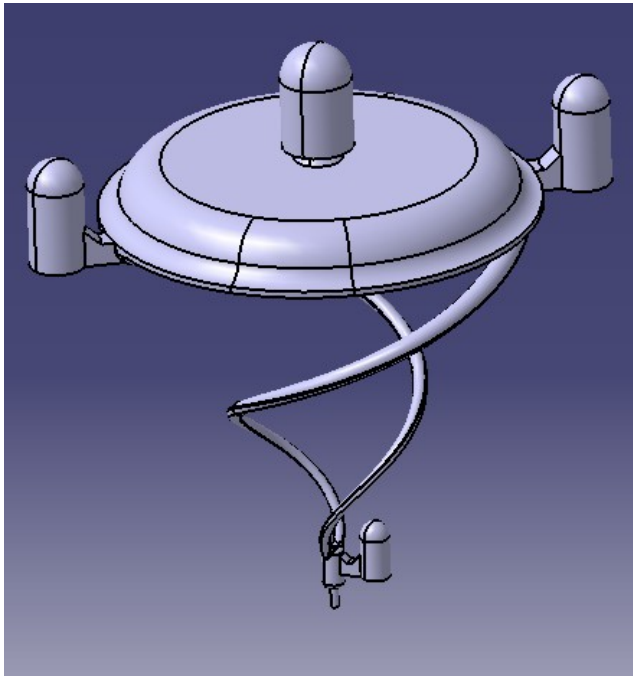
Positionnement

Retassures



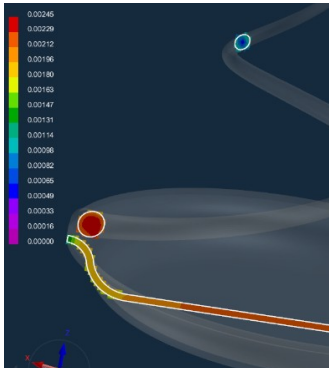
Points chauds



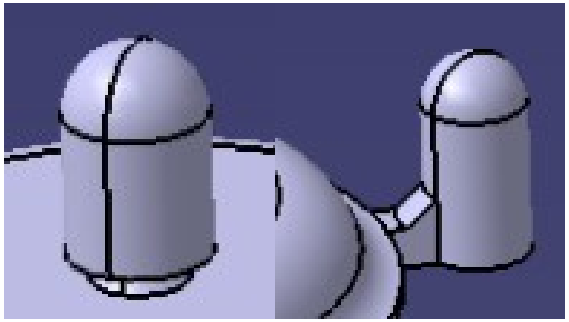


Dimensionnement des masselottes

Règle des modules

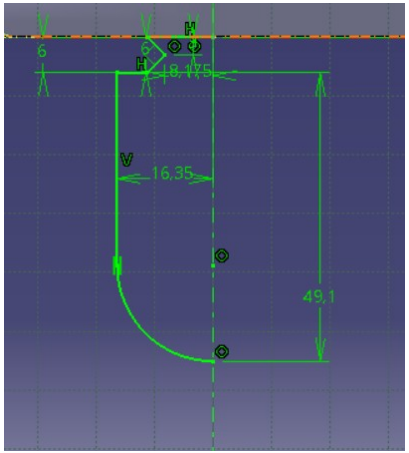


Quickcast → *M' (mm)*



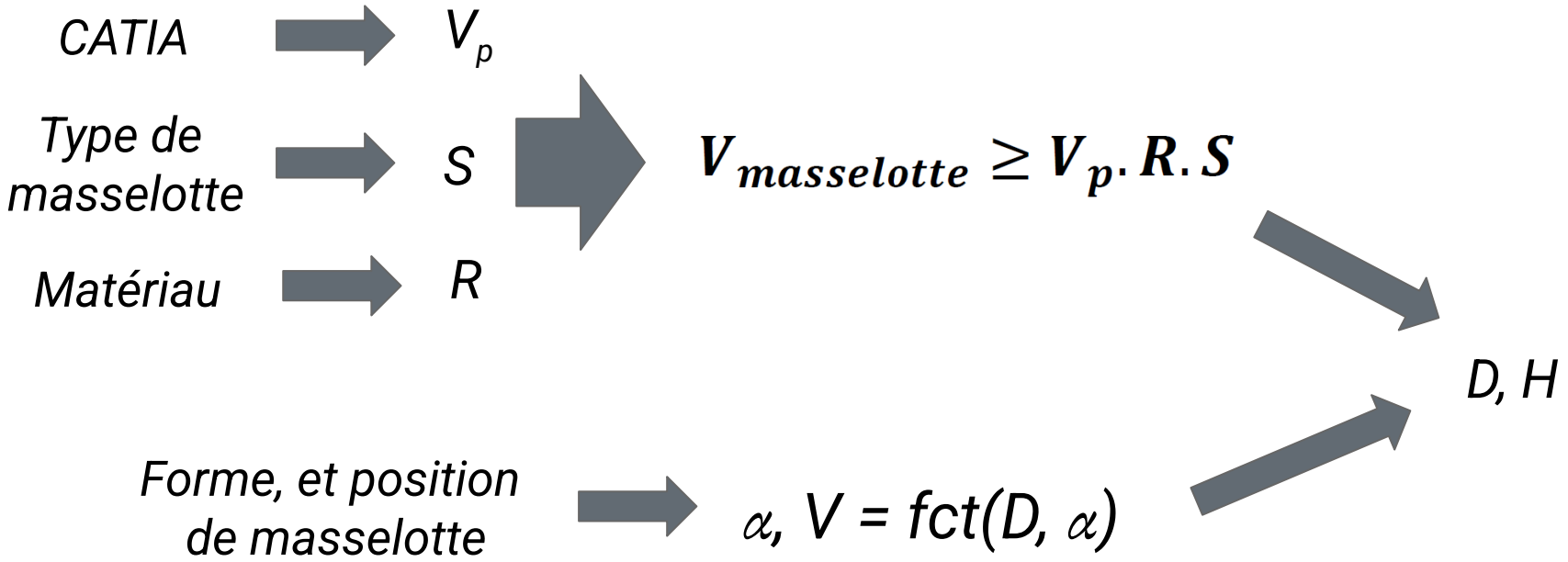
Position, type, forme de masselotte → J, Ω, α

Dimensions de la masselotte (D, H)



Dimensionnement des masselottes

Règle des retraits volumiques



Dimensionnement des masselottes

Choix final - Col de liaisons

Comparaison

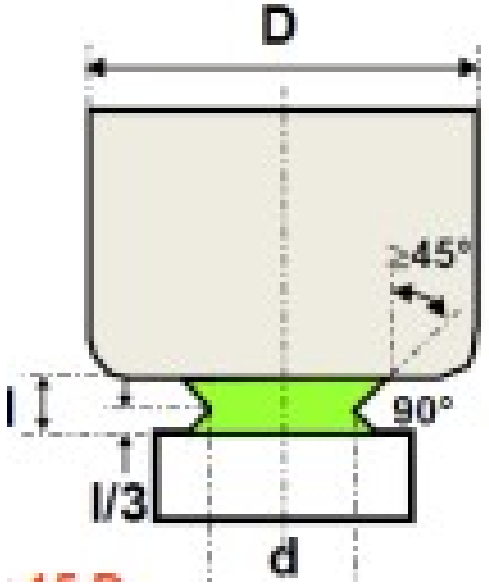
$V_{modules}$ *et* $V_{retraits\ volumiques}$

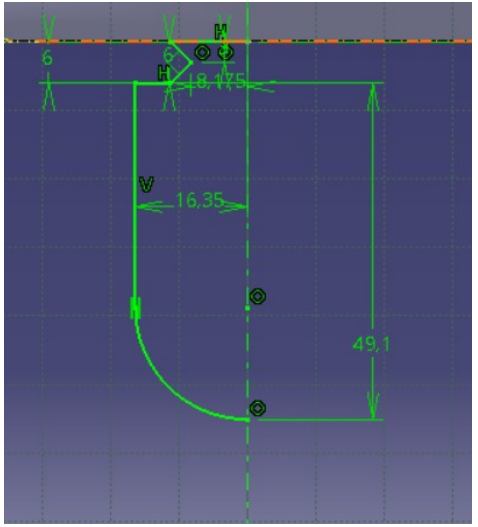
⬇

Choix conservatif

⬇

D, H finaux

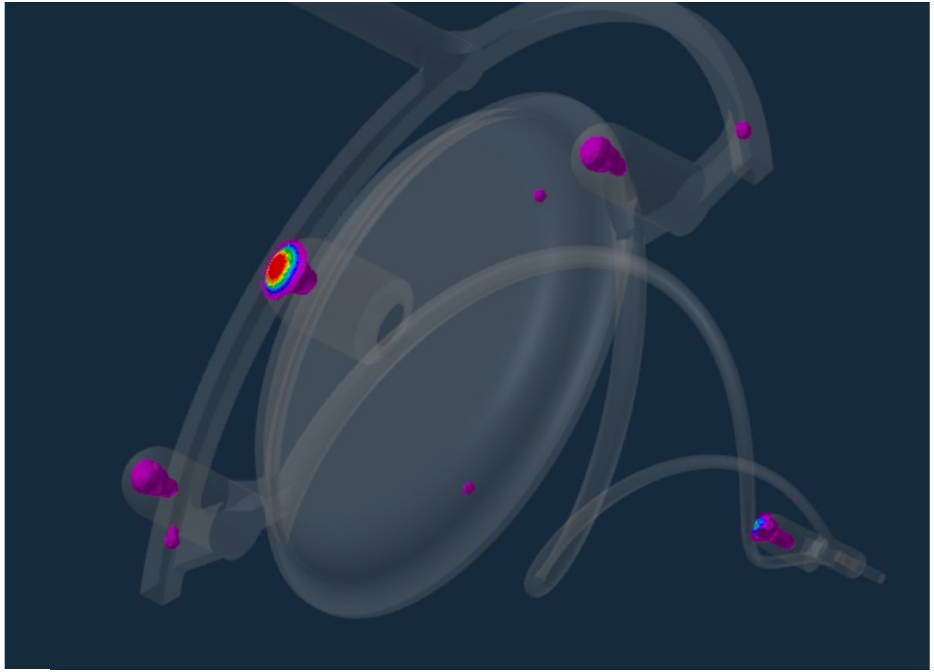




l > 5 mm et d = 0.5D

Dimensionnement des masselottes

Règle des rayons d'action



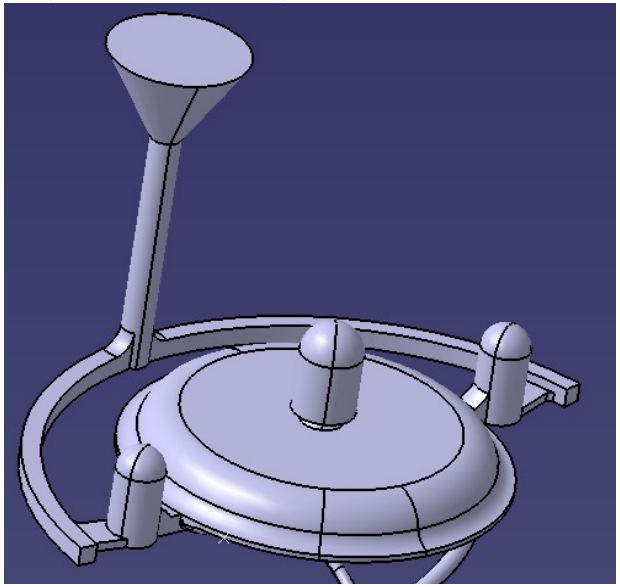
Dimensionnement du système de remplissage

Choix du remplissage - Épaisseur équivalente - Temps de remplissage

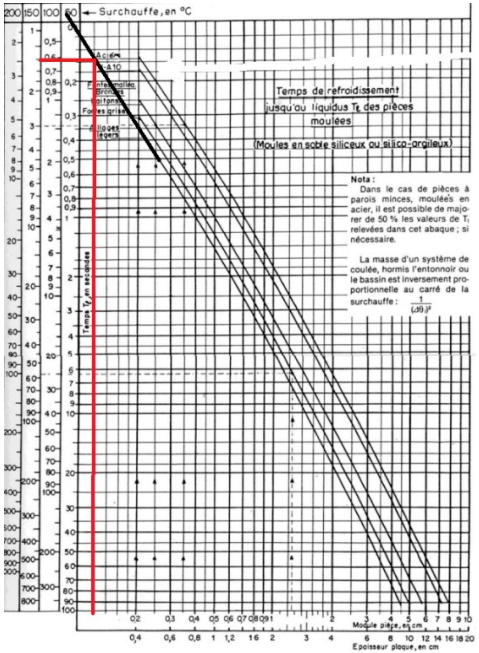


M' (mm)	1.16
e' (mm)	2.32

Remplissage latéral par les masselottes



t_r = 1,4s



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Dimensionnement du système de remplissage

Échelonnement - Rendement hydraulique

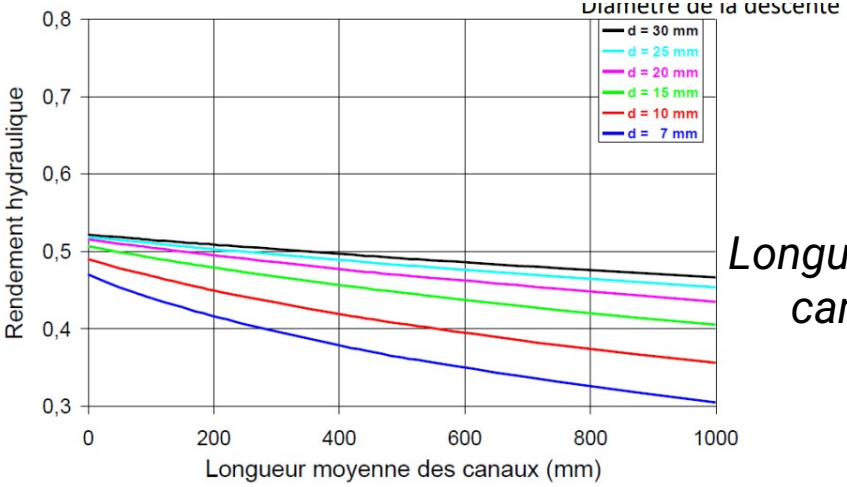
Matériau et hauteur métallostatique initiale

↓ *Oxydable, Hi > 100mm*

K et K'

↓

S_{min}



Longueur des canaux

R_h → S_{min} ↓ d_{min}

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Dimensionnement du système de remplissage

Sections - Modélisation - Vérification

S_{min}

↓

S_d, S_a, S_c

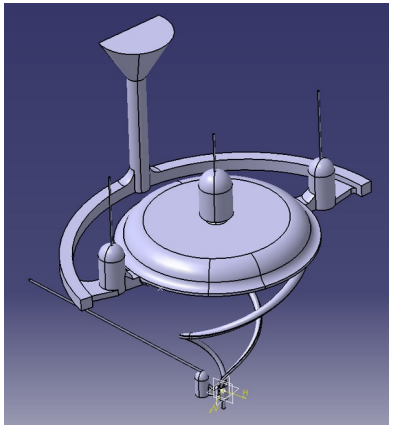
↓

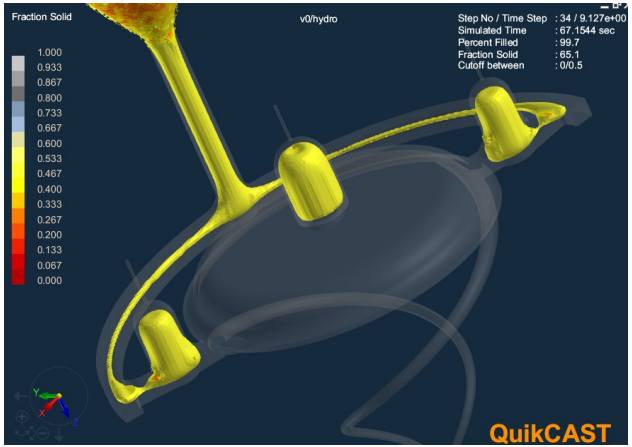
d_d, a, b, c, L_a

Attaques dans les masselottes

↓

$S_a < S_{cols}$

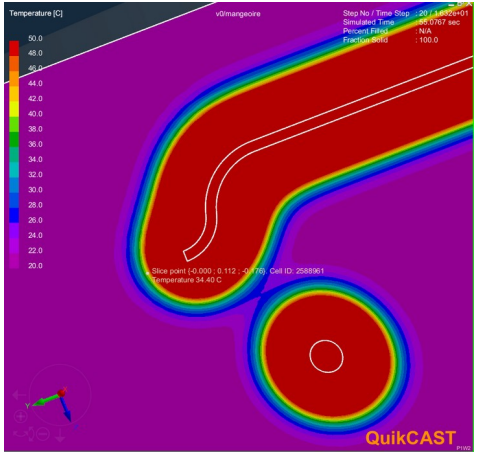


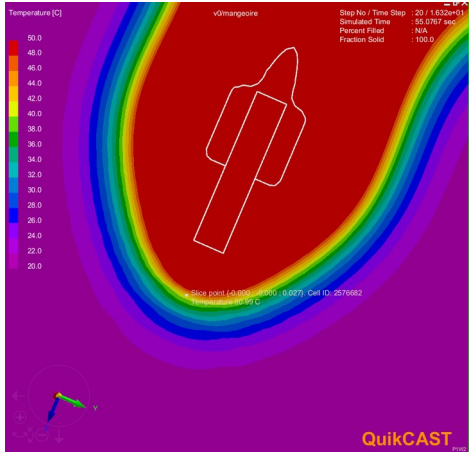


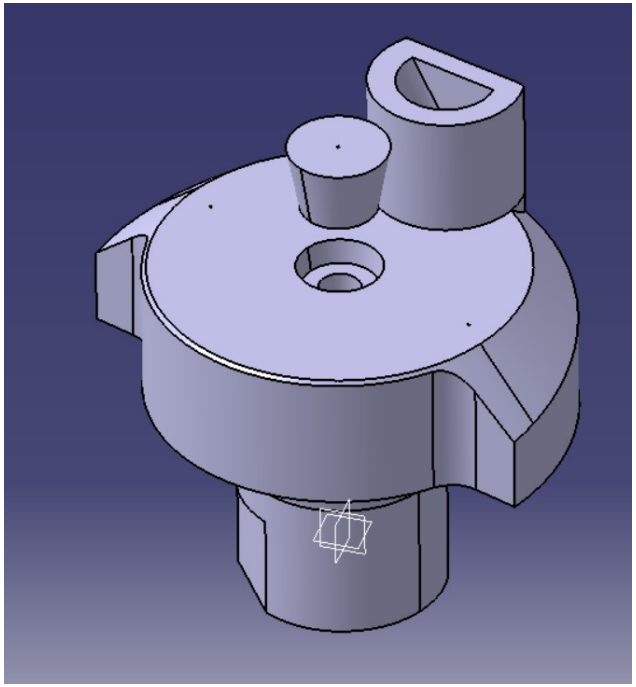
Modélisation du moule

Dimensionnement

T>50°C → *Moule*







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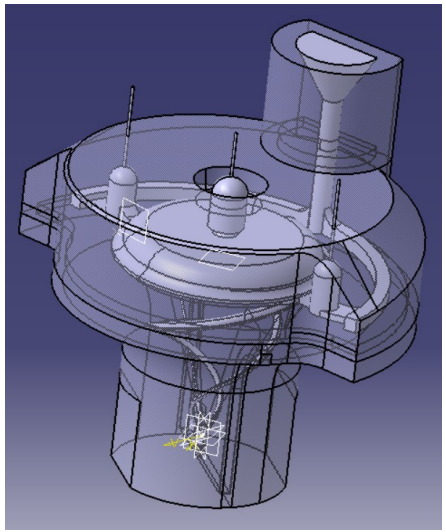
Mangeoire bas

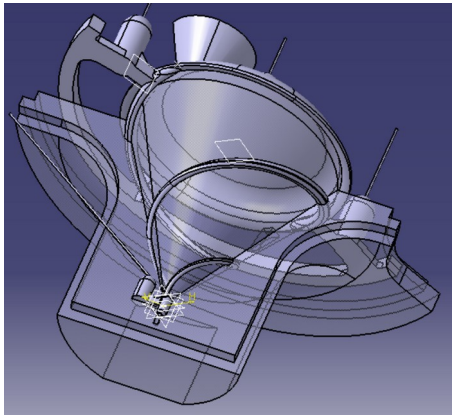
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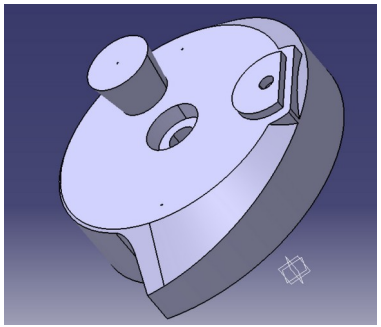
Modélisation du moule

Séparation des différentes parties du moule

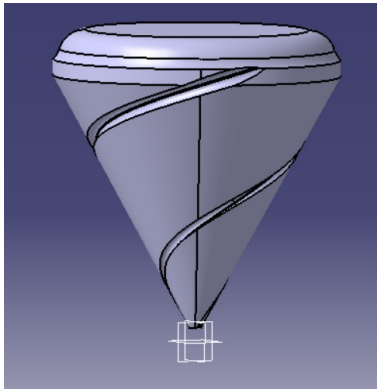
Découper pour nettoyer

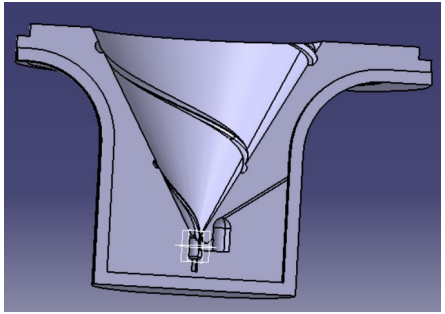












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Imp coulée





Flambage

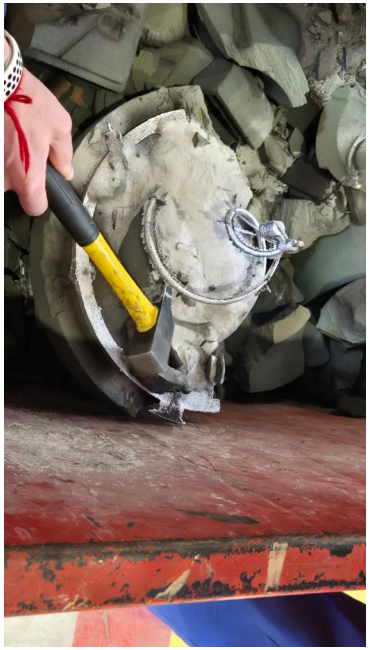


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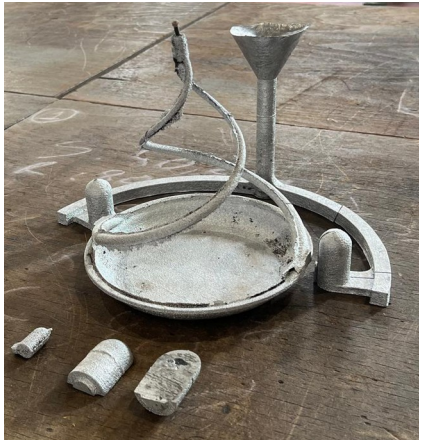
Impression du moule et coulée

Coulée de la pièce - Ébarbage - Mise au Mille









Mise au mille théorique : 1,6

Mise au mille expérimentale : 2,56

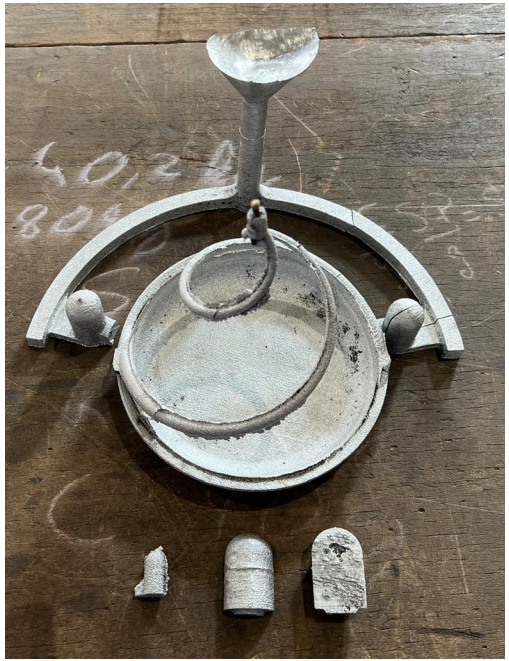
Pièce finale

Masselottes - Défauts









Finalisation

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17/17
Mangeoire bas

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